

What is the F in $F = ma$?

A three-case investigation based on *The Placeholder Force*

Working rule: In each case, separate the *empirical law that supplies the force* from Newton’s dynamical bridge, $\vec{F}_{\text{net}} = m_{\text{i}}\vec{a}$.

Name: _____

Class: _____

Article: <https://leelemacjames.github.io/placeHolderForce.pdf>

1. Coulomb force: what couples to what?

A fixed positive point charge $q_1 = +2.0\,\mu\text{C}$ is separated by $r = 0.30\,\text{m}$ from a second positive charge $q_2 = +1.0\,\mu\text{C}$. The second charge has inertial mass $m_{\text{i}} = 5.0\,\text{g}$. Ignore gravity and use $k_{\text{e}} = 9.0 \times 10^9\,\text{N m}^2\,\text{C}^{-2}$.

A. Represent the interaction

Draw the two charges. On q_2 , draw and label the force vector. Mark the *source charge*, the *coupling charge*, and the body whose inertia matters.

B. Join the two laws

Use

$$F_{\text{C}} = k_{\text{e}} \frac{|q_1 q_2|}{r^2}, \qquad F_{\text{net}} = m_{\text{i}} a$$

to find the initial force and acceleration of q_2 . Keep the two equations conceptually separate before combining them.

C. Interrogate the placeholder

What does Coulomb’s law contribute that $F = ma$ does not? Which property controls the coupling, and which controls resistance to acceleration? Would $F = ma$ alone tell you whether the interaction is attractive or repulsive?

D. Change one ratio

Without recalculating from the beginning, predict what happens to a if q_2/m_{i} doubles. Explain in one sentence.

2. Magnetic Lorentz force: force without speeding up?

An idealised positive particle has $q = +0.020\,\text{C}$, $m_{\text{i}} = 0.010\,\text{kg}$ and velocity $v = 5.0\,\text{m s}^{-1}$ to the right. It enters a uniform magnetic field $B = 0.20\,\text{T}$ directed into the page. Ignore all other forces.

A. Represent the interaction

Draw $\vec{v}$, $\vec{B}$ and the magnetic force $\vec{F}_B$ at entry. State how the direction was determined and identify the property that couples the particle to the field.

B. Join the two laws

Use

$$F_B = qvB, \qquad F_{\text{net}} = m_{\text{i}} a$$

to find the initial force and acceleration. Then use $qvB = m_{\text{i}} v^2/r$ to find the radius of curvature.

C. Interrogate the placeholder

Here F is perpendicular to the velocity. Explain how the particle can have a non-zero force and acceleration while its speed remains constant. Which part of the model tells you the force is perpendicular? Does $F = ma$ contain that fact?

D. Reverse the coupling

Replace the particle by one with charge $-q$ but the same mass and speed. State what changes and what remains unchanged.

3. Horizontal launch: motion where no force acts

A $0.50\,\text{kg}$ ball is launched horizontally at $8.0\,\text{m s}^{-1}$ from a height of $5.0\,\text{m}$ near Earth’s surface. Ignore air resistance and use $g = 9.8\,\text{m s}^{-2}$.

A. Represent the motion and force

Sketch the trajectory. At two points after release, draw the velocity and the *net force*. Make clear which quantities are horizontal and which are vertical.

B. Separate the components

Complete and explain:

$$F_x = \text{_____}, \quad a_x = \text{_____}, \quad v_x(t) = \text{_____}$$

$$F_y = m_{\text{g}} g = m_{\text{i}} a_y.$$

Using the usual identification $m_{\text{g}} = m_{\text{i}}$, find a_y , the time of flight and the horizontal range.

C. Interrogate the placeholder

Why does “no horizontal force” not mean “no horizontal motion”? What empirical claim is hidden when the same symbol m is used in both $F_{\text{g}} = mg$ and $F = ma$? Distinguish that claim from the approximation that Earth’s motion is neglected.

D. Remove the mass

A second ball has twice the inertial mass and is launched identically. Predict its trajectory under the stated assumptions and explain the cancellation involved.

Synthesis: what survives when the placeholder is opened?

1. Complete the comparison.

Case	Law supplying F	Coupling property versus inertial property
Coulomb		
Magnetic		
Projectile		

2. Develop the central claim.

Write a short paragraph responding to:

“ $F = ma$ is a dynamical bridge, not a description of the interaction.”

Use at least two cases. Refer to *source*, *coupling*, *inertial mass*, *direction* and *empirical law*. Then add one sentence explaining where a constitutive law such as Hooke’s law, $F = -kx$, would occupy the same architecture.

Teacher guide and indicative answers

What is the F in $F = ma$?

1. Coulomb force

Direction. Both charges are positive, so the force on q_2 is away from q_1 . q_1 is the source charge; q_2 is the responding or coupling charge; m_i measures inertial resistance.

$$F_C = 9.0 \times 10^9 \frac{(2.0 \times 10^{-6})(1.0 \times 10^{-6})}{(0.30)^2} = 0.20 \text{ N}$$

$$a = \frac{F_C}{m_i} = \frac{0.20}{5.0 \times 10^{-3}} = 40 \text{ m s}^{-2}.$$

Conceptual answer. Coulomb’s law supplies the inverse-square dependence, the dependence on the two charges, and attraction or repulsion through their signs. Newton’s second law converts the resulting net force into acceleration. It supplies none of those electrostatic details.

Ratio change. Since

$$a = k_e \frac{|q_1|}{r^2} \frac{|q_2|}{m_i},$$

doubling q_2/m_i doubles the initial acceleration.

Diagnostic misconception. Students may identify q_1 alone as “the force”. Press the distinction between a source property, a coupling property, the field or interaction law, and the force on the selected body.

2. Magnetic Lorentz force

For a positive charge, $\vec{F}_B = q\vec{v} \times \vec{B}$. With $\vec{v}$ rightwards and $\vec{B}$ into the page, the force is upwards.

$$F_B = qvB = (0.020)(5.0)(0.20) = 0.020 \text{ N}$$
$$a = \frac{0.020}{0.010} = 2.0 \text{ m s}^{-2}, \quad r = \frac{m_i v}{qB} = \frac{(0.010)(5.0)}{(0.020)(0.20)} = 12.5 \text{ m}.$$

Why speed is constant. The magnetic force is perpendicular to the instantaneous velocity, so it changes the direction of $\vec{v}$ rather than its magnitude. The cross product in the Lorentz law supplies this geometry; $F = ma$ alone does not.

Negative charge. The force and curvature reverse direction. The force magnitude, acceleration magnitude, speed and radius magnitude remain unchanged.

Diagnostic misconception. “Acceleration” is often read as “speeding up”. This case isolates vector acceleration as a change in velocity direction.

3. Horizontal projectile

After release, gravity is the only force under the stated model. The force is vertical and downward throughout; the velocity is tangent to the curved path.

$$F_x = 0, \quad a_x = 0, \quad v_x(t) = 8.0 \text{ m s}^{-1}.$$

Vertically,

$$m_g g = m_i a_y \quad \Rightarrow \quad a_y = \frac{m_g}{m_i} g.$$

With the empirical identification $m_g = m_i$, $a_y = g$ downward.

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{10}{9.8}} \approx 1.01 \text{ s}, \quad x = v_x t \approx 8.08 \text{ m}.$$

Conceptual answer. Zero horizontal force means zero horizontal acceleration, not zero horizontal velocity. The equality of gravitational and inertial mass is an empirical universality. Treating Earth as fixed is a separate source/test-body or negligible-back-reaction approximation.

Double mass. The ideal trajectory is unchanged because the same mass factor appears in gravitational coupling and inertial resistance and cancels.

Synthesis and classroom use

Indicative synthesis.

In all three cases, $F_{\text{net}} = m_i a$ performs the same operation: it maps a net interaction into acceleration. The physical provenance of the force comes from a separate empirical law. Coulomb’s law supplies source charge, coupling charge, inverse-square magnitude and sign dependence. The Lorentz law supplies charge, velocity and magnetic-field dependence together with the perpendicular direction. Near Earth, gravitation supplies $F_g = m_g g$, while equivalence permits m_g/m_i to disappear from the acceleration. The placeholder F therefore compresses distinct mechanisms into the same dynamical slot.

Discussion prompts.

- Which equation in each case is empirical closure, and which is dynamical balance?
- Does the named force describe a new force in addition to ma , or the interaction that constitutes the net force?
- Which properties belong to the source, the moving body, and the environment?
- What is gained and lost when all three are rewritten simply as $F = ma$?
- Hooke’s law is constitutive because it specifies a system-dependent response to deformation. Why can it still occupy the same left-hand slot as Coulomb or Lorentz force?

Suggested timing: 8 minutes per case, 10 minutes synthesis, 8 minutes whole-class comparison.